

BUNBURY, Australia

WMO: 946040

Lat: 33.3567S

Lon: 115.6447E

Elev: 6

StdP: 101.25

Time Zone: 8.00 (E08)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.1	4.2	1.5	4.2	6.9	2.6	4.6	7.7	11.5	14.8	10.5	14.9	0.6	90	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.5	33.8	19.9	32.0	19.5	30.4	19.2	21.9	29.0	21.1	28.0	20.4	27.4	4.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	14.7	24.2	19.0	13.8	23.5	18.0	13.0	23.0	64.2	29.3	61.2	27.9	58.6	27.3	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.5	0.8	37.5	1.1	1.4	0.1	38.5	-0.6	39.4	-1.2	40.2	-1.9	41.2		
(5)			0.4	23.3	1.0	1.1	-0.3	24.1	-0.9	24.8	-1.5	25.4	-2.3	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.8	22.2	22.3	20.6	17.7	14.9	13.0	12.1	12.4	13.1	15.0	18.2	20.1	(6)
(7)		DBStd	4.65	3.21	3.19	3.31	2.82	2.44	2.28	2.31	2.24	2.32	2.68	3.14	3.50	(7)
(8)		HDD10.0	17	0	0	0	0	1	2	7	4	3	0	0	0	(8)
(9)		HDD18.3	1042	5	4	16	45	109	159	192	186	157	108	41	19	(9)
(10)		CDD10.0	2487	378	346	329	230	152	93	73	77	96	156	245	314	(10)
(11)		CDD18.3	472	124	117	86	25	3	0	0	0	0	6	36	75	(11)
(12)		CDH23.3	4493	1284	1138	742	154	13	0	0	0	5	51	327	779	(12)
(13)	CDH26.7	1637	517	443	246	26	1	0	0	0	1	8	96	300	(13)	
(14)	Wind	WSAvg	3.7	4.5	4.3	3.9	3.1	2.9	3.1	3.4	3.3	3.7	3.8	4.2	4.2	(14)
(15)	Precipitation	PrecAvg	746	11	8	22	38	110	131	146	116	87	33	29	15	(15)
(16)		PrecMax	976	60	49	81	102	224	232	263	161	179	78	70	106	(16)
(17)		PrecMin	477	0	0	0	1	21	34	55	27	23	6	2	0	(17)
(18)		PrecStd	123	16	11	23	27	50	55	48	31	37	19	18	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.4	35.4	34.5	29.5	25.0	21.2	19.6	20.9	23.3	27.7	32.3	35.0	(19)
(20)			MCWB	20.1	20.8	20.3	18.3	16.1	14.6	14.9	15.0	15.1	17.8	18.5	19.4	(20)
(21)		2%	DB	33.5	33.3	31.5	27.1	22.7	19.8	18.4	19.0	20.4	24.6	29.5	32.3	(21)
(22)			MCWB	19.9	20.2	19.3	17.7	15.8	15.1	14.3	15.0	15.0	16.8	17.8	19.3	(22)
(23)		5%	DB	31.5	31.2	29.3	25.1	21.3	18.8	17.5	18.0	18.9	22.2	27.0	30.0	(23)
(24)			MCWB	19.7	19.5	18.7	17.4	15.9	14.8	13.9	14.4	14.6	16.1	17.5	18.7	(24)
(25)		10%	DB	29.5	29.5	27.3	23.3	20.2	17.9	16.7	17.0	17.8	20.4	24.8	27.9	(25)
(26)			MCWB	19.4	19.3	18.3	17.2	15.6	14.3	13.4	13.7	14.1	15.5	16.9	18.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.6	23.5	22.1	20.9	19.2	18.7	16.7	17.3	17.2	19.2	20.2	22.0	(27)
(28)			MCDWB	29.3	30.5	28.5	24.1	21.5	19.1	18.4	18.8	20.3	24.7	26.7	29.0	(28)
(29)		2%	WB	22.0	21.9	21.0	19.8	18.1	16.7	15.5	16.0	16.2	18.0	19.4	20.7	(29)
(30)			MCDWB	29.3	29.2	27.0	23.3	20.3	18.3	17.3	17.9	18.9	22.4	25.6	28.9	(30)
(31)		5%	WB	20.9	21.0	20.3	18.9	17.2	15.7	14.5	15.0	15.2	16.9	18.6	19.7	(31)
(32)			MCDWB	28.4	28.4	26.4	22.6	19.7	17.7	16.7	17.2	18.1	20.8	24.6	27.7	(32)
(33)		10%	WB	20.0	20.1	19.5	18.0	16.3	14.8	13.7	14.1	14.5	16.0	17.7	18.8	(33)
(34)			MCDWB	28.2	27.6	25.7	22.4	19.2	17.1	16.1	16.4	17.3	19.9	23.6	26.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.8	12.5	11.9	10.9	10.5	9.5	8.8	9.0	9.2	10.1	11.0	12.0	(35)
(36)			MCDBR	15.5	14.7	14.2	12.9	12.1	10.6	9.3	10.0	10.6	12.9	14.3	15.3	(36)
(37)			MCWBR	6.0	5.5	5.7	6.1	7.0	6.8	6.4	6.7	6.8	7.1	6.5	6.2	(37)
(38)		5% WB	MCDBR	12.8	12.3	11.2	9.6	9.0	8.0	7.4	8.0	8.9	10.5	11.8	13.4	(38)
(39)			MCWBR	6.2	5.6	5.6	5.9	6.6	6.4	6.0	6.3	6.5	6.7	6.4	6.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.332	0.334	0.321	0.314	0.309	0.308	0.304	0.313	0.328	0.325	0.331	0.327	(40)	
(41)		taud	2.588	2.588	2.626	2.640	2.625	2.604	2.614	2.576	2.513	2.538	2.546	2.577	(41)	
(42)		Ebn at Noon	1005	983	962	916	868	840	864	902	939	981	1001	1014	(42)	
(43)		Edh at Noon	105	101	91	80	72	69	72	84	100	105	108	106	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.96	7.09	5.72	4.07	2.95	2.38	2.54	3.36	4.51	5.98	7.23	7.99	(44)	
(45)		RadStd	0.41	0.28	0.28	0.25	0.20	0.15	0.14	0.20	0.26	0.29	0.42	0.40	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page